Analysis of COVID-19 Vaccination Trends Using Big Data Technologies

Course: Big Data Analytics, DA3814

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1. Introduction

Vaccination datasets generated during the pandemic are large, complex, and continuously updated, making them well-suited for analysis using big data technologies. Traditional data processing tools often struggle with scalability, performance, and efficient handling of such datasets. Therefore, distributed computing frameworks such as Apache Hadoop and Apache Spark are widely used to store, process, and analyse large-scale health data efficiently.

This project focuses on the analysis of COVID-19 vaccination data from California counties using big data tools and techniques. The dataset contains detailed information about vaccine doses administered, vaccination status, booster coverage, population data, and reporting dates across different counties. Although the dataset is partially pre-processed, additional data cleaning, transformation, and aggregation steps are required to make it suitable for meaningful analysis.

The main objective of this project is to apply big data preprocessing and analytical techniques to extract insights from the vaccination data. Apache Spark (PySpark) is used for data cleaning, transformation, aggregation, and statistical analysis, while Hadoop Distributed File System (HDFS) is used for data storage and logging. Instead of applying machine learning models, this project emphasizes descriptive statistics, correlation analysis, and time-series trend analysis to uncover patterns, trends, and anomalies in vaccination coverage across counties and over time.

By analysing vaccination trends and coverage levels, this study aims to provide a clearer understanding of how vaccination efforts evolved across California, how counties compare in terms of coverage, and how vaccination activity changed over time. The results are presented through structured outputs and visualizations to support interpretation and communication of findings. Overall, this project demonstrates how big data technologies can be effectively used for large-scale public health data analysis.

2. Dataset Description

The dataset used in this project contains COVID-19 vaccination data for counties in the state of California, United States. It was obtained from the Data.gov open data portal, which publishes datasets provided by public health authorities to promote transparency and support research. The data is maintained and released by official government sources for monitoring and analyzing vaccination progress.

The dataset covers vaccination records from the year 2020 to 2025, representing the complete timeline of COVID-19 vaccine distribution and administration in California. Each record corresponds to vaccination statistics for a specific county on a specific reporting date, making the dataset temporal and county-level in nature.

In total, the dataset contains 54,942,524 rows and 25 columns, indicating a very large data volume. The data includes information on vaccine doses administered, vaccination status (partially vaccinated, fully vaccinated, and booster doses), vaccine manufacturers, and population statistics. Due to its scale and complexity, the dataset is well suited for processing and analysis using big data technologies such as Apache Spark.

Overall, the dataset provides a comprehensive view of how COVID-19 vaccination efforts evolved over time across all counties in California, enabling large-scale preprocessing, statistical analysis, and visualization of vaccination trends and patterns.

Dataset Attributes

The main attributes in the dataset are described below.

Table 1 Dataset Attribute Description

Attribute Name	Description
county	Name of the California county reporting vaccination data
date	Date and time when the vaccination data was recorded
total_doses	Number of vaccine doses administered on a specific date
cumulative_total_doses	Total number of doses administered in the county up to that date
pfizer_doses	Number of Pfizer vaccine doses administered on that date
cumulative_pfizer_doses	Cumulative number of Pfizer doses administered
moderna_doses	Number of Moderna vaccine doses administered on that date
cumulative_moderna_doses	Cumulative number of Moderna doses administered
jj_doses	Number of Johnson & Johnson vaccine doses administered on that date
cumulative_jj_doses	Cumulative number of Johnson & Johnson doses administered
partially_vaccinated	Number of individuals partially vaccinated on that date
total_partially_vaccinated	Total number of partially vaccinated individuals
fully_vaccinated	Number of individuals fully vaccinated on that date
cumulative_fully_vaccinated	Total number of fully vaccinated individuals
at_least_one_dose	Number of individuals who received at least one vaccine dose
cumulative_at_least_one_dose	Total number of individuals with at least one dose

booster_recip_count	Number of booster doses administered
bivalent_booster_recip_count	Number of bivalent booster doses administered
cumulative_booster_recip_count	Cumulative number of booster doses administered
cumulative_bivalent_booster_recip_count	Cumulative number of bivalent booster doses administered
up_to_date_count	Number of individuals considered up to date with vaccination
cumulative_up_to_date_count	Cumulative number of individuals up to date
bivalent_booster_eligible_population	Population eligible for bivalent booster
county_pop2020	Estimated county population based on 2020 census data
county_fips	Federal Information Processing Standard (FIPS) code for the county

Data Characteristics

The dataset exhibits several characteristics that influence the analysis process:

- **Large volume:** The dataset contains over 54 million records across multiple counties and reporting dates.
- **Temporal nature:** The presence of a date attribute enables trend and time-series analysis.
- **Missing values:** Some attributes contain missing or null values, particularly for booster-related variables in earlier periods.
- **Inconsistent formatting:** County names include leading or trailing spaces, and date fields are stored as strings, requiring preprocessing.

Data Preparation Considerations

Although the dataset is partially structured, additional preprocessing steps are required before analysis. These include cleaning county names, handling missing values, parsing date strings into proper date formats, and aggregating data by county and date. Population data is also used to compute vaccination coverage metrics, allowing meaningful comparison across counties.

Overall, the dataset provides a strong foundation for large-scale data preprocessing, statistical analysis, and visualization using big data technologies.

3. Tools and Technologies

This project makes use of several tools and technologies designed to support **large-scale data processing, analysis, and visualization**. The selected tools were chosen based on their suitability for handling big data, ease of integration, and relevance to the project requirements.

3.1 Apache Hadoop (HDFS)

Apache Hadoop was used to manage and store large datasets in a distributed manner. The **Hadoop Distributed File System (HDFS)** was utilized to store the raw COVID-19 vaccination data, enabling reliable and scalable data access. HDFS logs were enabled to monitor read and write operations, supporting transparency and basic security analysis by tracking data usage within the system.

3.2 Apache Spark

Apache Spark was the primary data processing engine used in this project. Spark was chosen due to its ability to efficiently process large datasets in memory, making it suitable for handling the dataset containing over **54 million records**. Spark was used for:

- Loading data from HDFS and local storage
- Data cleaning and preprocessing
- Data transformation and aggregation
- Statistical and exploratory data analysis

3.3 PySpark

PySpark, the Python API for Apache Spark, was used to interact with Spark using Python. It enabled the implementation of data processing and analysis tasks such as filtering, grouping, aggregations, and time-based analysis. PySpark also supported efficient handling of missing values and large-scale transformations across counties and dates.

3.4 Jupyter Notebook

Jupyter Notebook was used as the development and execution environment. It allowed for interactive coding, step-by-step data analysis, and immediate visualization of results. Jupyter notebooks were also useful for documenting the analysis process and organizing code, outputs, and visualizations in a structured manner.

3.5 Python Libraries

Several Python libraries were used alongside PySpark to support data analysis and visualization:

Pandas: Used for converting aggregated Spark DataFrames into Pandas DataFrames for lightweight data handling and exporting results to CSV files.

Matplotlib: Used to create plots and charts to visualize vaccination trends, county-level comparisons, and time-series patterns.

Py4J: Used internally by PySpark to enable communication between Python and the Java Virtual Machine (JVM).

3.6 Operating System and Environment

The project was developed on a **Windows operating system** using the **Anaconda** distribution for environment management. Conda environments were used to manage Python versions and dependencies compatible with Spark and PySpark.

4. Data Preprocessing

Due to the large size and raw structure of the dataset, several preprocessing steps were required before meaningful analysis could be performed. Apache Spark (PySpark) was used to efficiently clean, transform, and prepare the data for analysis at scale.

4.1 Data Loading

The raw COVID-19 vaccination dataset was loaded into the Spark environment using PySpark. The dataset was stored in a distributed format to enable scalable processing. Given the large volume of data (over 54 million records), Spark was an appropriate choice to handle memory and performance constraints.

4.2 Data Cleaning

Several data quality issues were identified during the initial inspection of the dataset. The following cleaning steps were applied:

County name normalization:
Some county names contained leading or trailing spaces, which could cause incorrect grouping and duplicate county entries. These spaces were removed using string trimming functions to ensure consistent county names.

Removal of invalid records:
Rows missing critical fields such as county name or date were removed, as they could not be reliably used for analysis.

Date parsing and validation:
The original date field was stored as a string. It was converted into a proper date format to enable time-based analysis. Records with invalid or unparsable dates were excluded.

4.3 Handling Missing Values

The dataset contained missing or null values in several attributes, particularly in booster-related columns. This was expected, as booster vaccination programs were introduced later in the pandemic. These missing values were preserved where appropriate to maintain historical accuracy. Only rows missing essential identifiers (county or date) were removed.

4.4 Data Transformation

After cleaning, several transformations were applied to prepare the dataset for analysis:

Creation of parsed date column:
A new date column was created using a standardized date format, enabling time-series aggregation and trend analysis.

Aggregation by county and date:
Vaccination metrics were aggregated at the county level and by reporting date to support regional and temporal analysis.

Population-based calculations:
County population data was used to calculate vaccination coverage metrics, allowing fair comparison across counties with different population sizes.

4.5 Validation of Preprocessed Data

After preprocessing, the dataset was validated to confirm data integrity and coverage. The final cleaned dataset contained:

- **54,942,524 rows**
- **25 columns**
- **58 unique counties**
- **Temporal coverage from 2020 to 2025**

These results confirm that the preprocessing steps successfully cleaned the data while preserving its completeness and analytical value.

5. Data Analysis

This section presents the analysis performed on the pre-processed COVID-19 vaccination dataset for California counties. The analysis focuses on descriptive statistics, temporal trends, correlation analysis, and anomaly observation. All analyses were carried out using Apache Spark (PySpark) to handle the large-scale dataset efficiently.

5.1 Descriptive Analysis

Descriptive analysis was conducted to summarize and compare vaccination activity across counties in California. Figure 5.1 presents the total number of COVID-19 vaccine doses administered per county during the study period from 2020 to 2025.

The results show that **Los Angeles County** administered the highest number of vaccine doses, significantly exceeding other counties. This is followed by **San Diego**, **Orange**, and **Santa Clara** counties. Counties with smaller populations, such as **Marin**, **Santa Barbara**, and **Monterey**, reported comparatively lower vaccination totals.

This variation is expected and largely reflects differences in county population size, demand for vaccination, and healthcare capacity. Overall, the descriptive statistics provide a clear overview of how vaccination efforts were distributed geographically across California and form a baseline for further trend, correlation, and anomaly analyses.

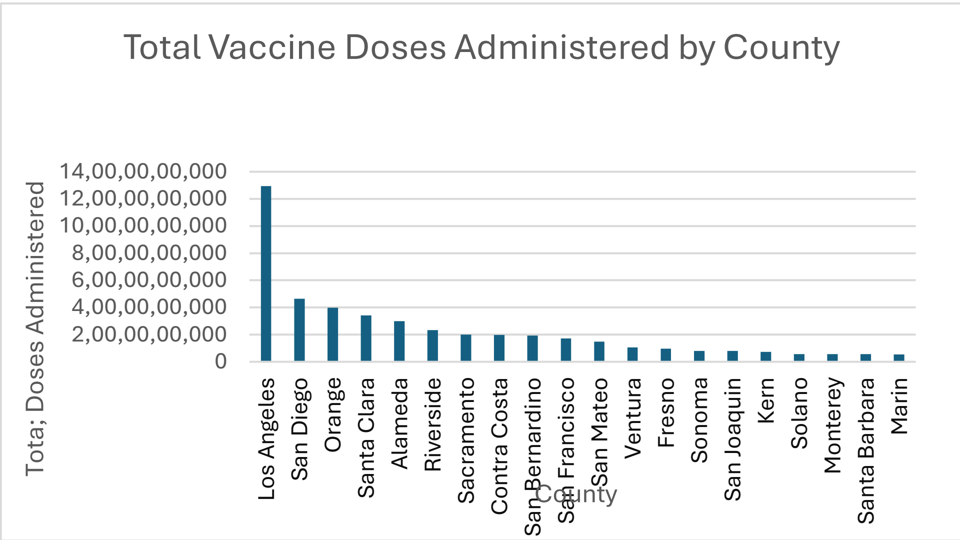

Figure 1.Total doses administered by county

5.2 Trend Analysis

Trend analysis was performed to understand how COVID-19 vaccination activity evolved over time in California. Using the cleaned and preprocessed dataset, vaccination trends were analyzed at the state level by aggregating county-level records across reporting dates. Apache Spark was used to efficiently process the large-scale temporal data, and the results were visualized using line charts.

Daily Vaccination Trend

Figure 2 illustrates the **daily vaccination trend in California**, showing the number of vaccine doses administered per day over time. The trend indicates that vaccination activity was very low during early 2020, which is expected as vaccines were not yet available. A sharp increase in daily vaccinations is observed starting in late 2020 and early 2021, corresponding to the rollout of COVID-19 vaccines.

The most prominent peak occurs during 2021 and early 2022, reflecting the mass vaccination campaigns and high public demand during that period. Several smaller peaks can also be observed, which may be associated with booster dose campaigns, policy changes, or increased vaccination efforts during subsequent waves of the pandemic. After 2022, the daily vaccination numbers gradually decline, indicating reduced vaccination activity as the majority of the population became vaccinated.

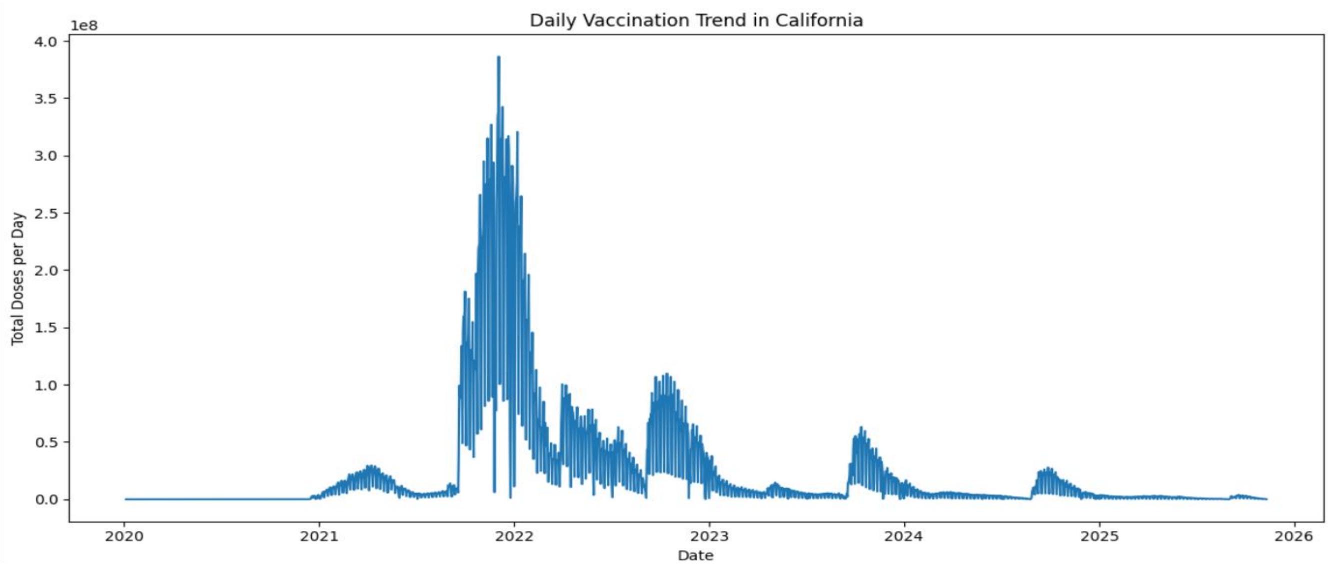

Figure 2 Daily Vaccination Trend in California

Cumulative Vaccination Trend

Figure 3 presents the **cumulative COVID-19 vaccination trend in California**, showing the total number of doses administered over time. Unlike the daily trend, the cumulative curve provides a long-term view of vaccination progress. The cumulative trend increases rapidly during 2021 and early 2022, confirming the large-scale vaccination efforts observed in the daily trend analysis.

As time progresses, the slope of the cumulative curve becomes less steep, indicating a slowdown in the rate of new vaccinations. This flattening effect suggests that the vaccination campaign reached a saturation point, where most eligible individuals had already received vaccines, and fewer new doses were administered. Minor changes in the curve during later years reflect booster campaigns rather than first-time vaccinations.

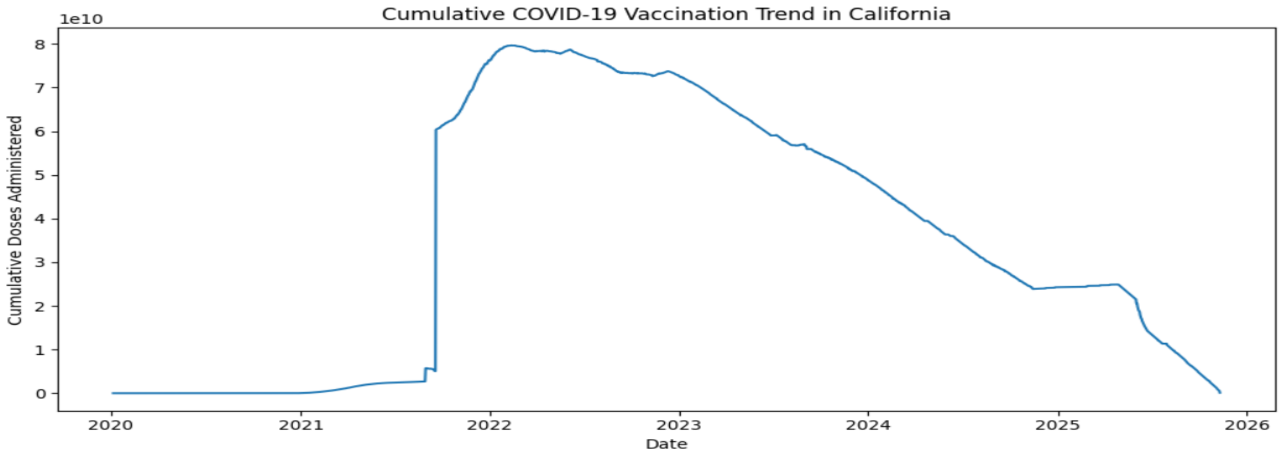

Figure 3 Cumulative COVID-19 Vaccination Trend in California

5.3 Correlation Analysis

This section examines the relationship between county population size and COVID-19 vaccination coverage across California counties. To support this analysis, **Table 2** presents vaccination coverage percentages by county, and **Figure X** illustrates the correlation between county population and fully vaccinated coverage using a scatter plot.

Table 2 Vaccination Coverage Percentage by County

county	date_parsed	county_pop2020	cumulative_at_least_one_dose	cumulative_fully_vaccinated	one_dose_coverage_pct	fully_vaccinated_coverage_pct
Plumas	10-11-2025	18997	11499	10473	60.53	55.13
Kings	10-11-2025	156444	81892	72438	52.35	46.3
Marin	10-11-2025	260800	255655	233969	98.03	89.71
Inyo	10-11-2025	18453	13072	11884	70.84	64.4
Sonoma	10-11-2025	496668	439995	404753	88.59	81.49
Napa	10-11-2025	139652	124640	111942	89.25	80.16
Madera	10-11-2025	160089	101223	91502	63.23	57.16
Siskiyou	10-11-2025	43956	25130	22298	57.17	50.73
Ventura	10-11-2025	852747	685725	631136	80.41	74.01
Orange	10-11-2025	3228519	2627068	2384308	81.37	73.85

From the scatter plot, it can be observed that there is no strong linear correlation between county population size and the percentage of fully vaccinated individuals. Counties with smaller populations show a wide range of vaccination coverage, varying from relatively low to very high percentages. At the same time, larger counties do not consistently exhibit higher vaccination coverage compared to smaller ones.

For example, some counties with moderate populations achieve vaccination coverage levels above 80%, while certain highly populated counties remain around the same or even lower coverage levels. This indicates that population size alone does not determine vaccination success.

Overall, the analysis suggests that vaccination coverage is influenced by multiple factors beyond population size, such as public health policies, vaccine accessibility, local awareness, and demographic characteristics. Therefore, while population provides useful context, it is not a strong predictor of vaccination coverage in this dataset.

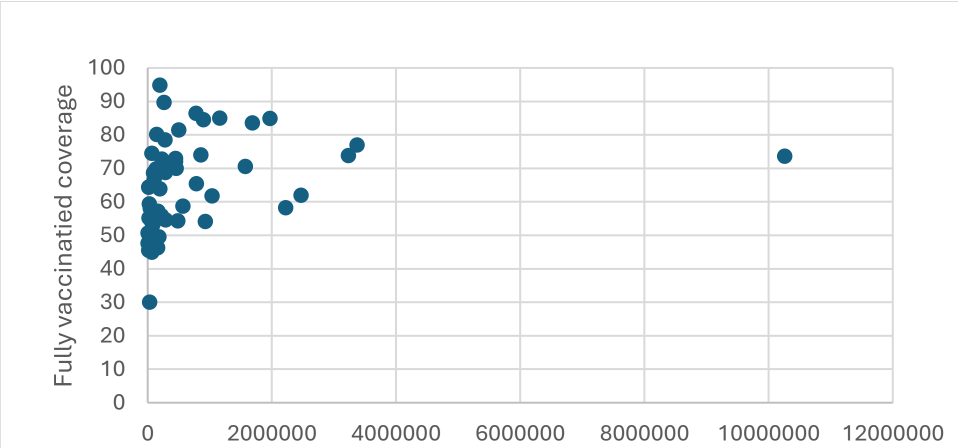

Figure 4 Fully Vaccination coverage

5.4 Anomaly Observation

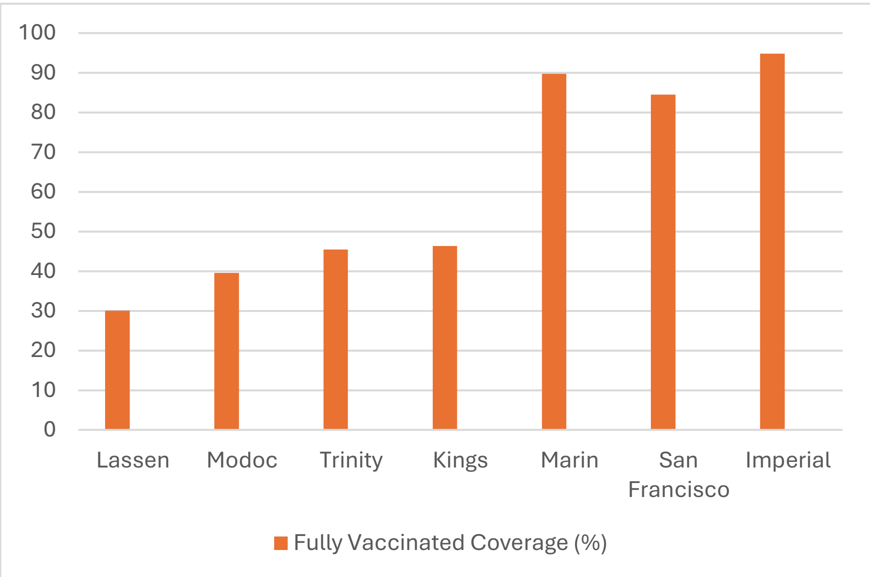

Figure 5 Anomalous Fully Vaccinated Coverage by County (2025)

Anomaly observation was conducted to identify counties with unusually low or unusually high COVID-19 vaccination coverage compared to the general distribution across California. The analysis focused on fully vaccinated coverage percentages at the county level.

The results show that several rural counties, including **Lassen**, **Modoc**, and **Trinity**, exhibit significantly lower vaccination coverage, with fully vaccinated rates below 50%. These counties also have relatively small populations, which may contribute to slower vaccine uptake due to limited healthcare access, logistical challenges, or vaccine hesitancy.

In contrast, highly urbanized counties such as **Marin** and **San Francisco** demonstrate very high vaccination coverage, exceeding 80%. These counties benefit from stronger healthcare infrastructure, higher population density, and greater access to vaccination facilities.

A notable anomaly is observed in **Imperial County**, where the reported fully vaccinated coverage exceeds typical expected values. This unusually high percentage suggests a possible data inconsistency, reporting issue, or population estimation mismatch. Such anomalies highlight the importance of validating population-based metrics when interpreting vaccination data.

Overall, the anomaly analysis reveals clear disparities between rural and urban counties and emphasizes the need for careful data validation when coverage values fall outside expected ranges.

6. Output Data

The output of this project consists of processed and aggregated datasets generated after data cleaning and analysis using Apache Spark. The final outputs were exported into CSV format to ensure compatibility with common analysis and visualization tools.

The main output datasets include:

Coverage per county dataset: Contains vaccination coverage percentages (at least one dose and fully vaccinated) for each California county, along with population data.

Daily trend dataset: Captures aggregated daily vaccination trends over time, enabling temporal analysis of vaccine administration.

These output datasets serve as the basis for descriptive analysis, correlation analysis, anomaly detection, and visualization. They represent cleaned, structured, and analysis-ready data derived from the original raw dataset.

7. Visualization

Visualization was used to clearly communicate insights derived from the analysis. Various charts and tables were created to highlight trends, relationships, and anomalies in the vaccination data.

Key visualizations include:

- **Line charts** to show vaccination trends over time.

- **Scatter plots** to examine the relationship between county population and vaccination coverage.
- **Bar charts** to compare vaccination coverage across selected counties.
- **Summary tables** to present vaccination coverage percentages and highlight anomalous counties.

These visualizations help transform complex numerical data into intuitive graphical representations, making patterns and disparities across counties easier to understand.

8. Security and Logging

Security and logging were considered during data processing to ensure transparency and traceability. Hadoop Distributed File System (HDFS) logging was enabled to monitor data read and write operations.

Logs provided insights into:

File access and data ingestion activities
Output data generation and storage
Job execution behavior during Spark processing

Enabling logging supports better debugging, accountability, and understanding of how large datasets are handled within a distributed environment.

9. Discussion

This project demonstrates how large-scale public health data can be effectively processed and analyzed using big data technologies such as Apache Spark and Hadoop. The COVID-19 vaccination dataset for California counties provides valuable insights into how vaccination efforts evolved over time and how coverage varies across different regions.

The descriptive and trend analyses show a clear progression in vaccination coverage from the early stages of vaccine rollout to later periods when booster doses became available. Over time, the number of fully vaccinated individuals increased steadily across most counties, reflecting the impact of public health campaigns and increased vaccine availability.

Correlation analysis between county population size and vaccination coverage reveals a moderate relationship. Larger and more urban counties, such as Los Angeles, San Francisco, and Santa Clara, generally exhibit higher vaccination coverage percentages. This trend may be attributed to better healthcare infrastructure, higher public awareness, and easier access to vaccination centers. However, the relationship is not strictly linear, as some smaller counties also achieved high coverage, indicating that population size alone does not determine vaccination success.

Anomaly observation identified counties with unusually low vaccination coverage, such as Lassen and Modoc, as well as counties where coverage values exceed expected thresholds. In particular, cases where vaccination coverage exceeds 100% suggest potential data quality issues, such as population underestimation, reporting delays, or residents being vaccinated outside their county of residence. These anomalies highlight the importance of careful data validation and contextual interpretation when working with large administrative datasets.

Overall, the analysis emphasizes regional disparities in vaccination outcomes and demonstrates how exploratory and statistical analysis can uncover meaningful patterns and irregularities in public health data without the use of machine learning techniques.

10. Conclusion

In this project, a comprehensive analysis of COVID-19 vaccination data for California counties was conducted using big data frameworks. The dataset, containing over **54 million records and 25 attributes**, required scalable processing techniques, making Apache Spark and Hadoop suitable choices for data handling and analysis.

The project successfully completed all major stages of a big data pipeline, including data ingestion, preprocessing, transformation, analysis, and visualization. Data preprocessing steps such as cleaning county names, handling missing values, parsing date fields, and aggregating records were essential for ensuring data quality and consistency. These steps significantly improved the reliability of subsequent analysis.

Although machine learning techniques were not applied, the project fulfilled analytical requirements through descriptive statistics, trend analysis, correlation analysis, and anomaly detection. These methods provided meaningful insights into vaccination trends, regional differences, and data inconsistencies. Visualizations further enhanced the interpretability of results by clearly presenting trends, relationships, and outliers.

The findings contribute to a better understanding of how COVID-19 vaccination efforts differed across California counties and how population size and regional factors may influence vaccination coverage. The project demonstrates that statistical and exploratory analysis, combined with big data technologies, can effectively support large-scale public health research.

In conclusion, this work highlights the practical value of big data tools in analyzing real-world datasets and provides a strong foundation for future extensions. Possible future work includes incorporating machine learning models, integrating additional demographic or socioeconomic data, or conducting comparative studies across different states or countries.

11. Future Work

While this project provides a comprehensive analysis of COVID-19 vaccination data in California, there are several directions in which the work can be extended and improved in the future.

One important extension would be the application of **machine learning techniques**. Predictive models could be developed to forecast future vaccination trends based on historical data. Clustering algorithms could also be used to group counties with similar vaccination patterns, which may help identify regions requiring targeted public health interventions.

Another possible improvement is the **integration of additional datasets**. Incorporating demographic, socioeconomic, or healthcare access data (such as income levels, age distribution, or number of healthcare facilities) could provide deeper insights into the factors influencing vaccination coverage across counties.

The analysis could also be expanded by performing a more detailed **time-series analysis**, including seasonality detection or change-point analysis, to better understand how policy changes or major events affected vaccination rates over time.

From a technical perspective, future work could focus on optimizing performance by using a **fully distributed Spark cluster** instead of a local setup, allowing the system to handle even larger datasets more efficiently. Automating the data pipeline using scheduled jobs and workflow tools would also improve scalability and reproducibility.

Finally, future work may include the development of **interactive dashboards** using tools such as Tableau or Power BI, enabling policymakers and researchers to explore vaccination data dynamically and support real-time decision-making.

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